Field Case Study Report on Avare Biotech App Validation powered by MAKSA technology on Frozen-Thawed Bovine Semen (Punjab, Pakistan, 2026)

Date: MARCH 2026
Location: Punjab, Pakistan

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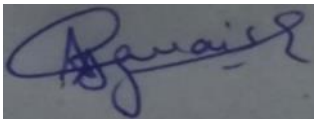
Signature: 	Signature:

Executive Summary / Conclusions

- The MAKSA technology showed convenient and effective performance when analyzing frozen–thawed Bovine semen samples.
- Total motility measurements obtained through MAKSA were quite robust.
- Progressive motility values recorded by MAKSA were generally higher, most likely due to the system's extra detectability of progressive motion of sperms in thawed samples.
- Accurate concentration analysis requires semen videos to be recorded exactly according to Avare's guidelines. The results are highly sensitive to recording quality, particularly magnification. Insufficient zoom and differences in cameras or microscopes are the primary reasons for variation in concentration values.
- The graphical demonstration of the results in a Pie chart in MAKSA report are a plus point.
- Video capture procedures followed AVARE guidelines, lead to most consistent and effective results

Profit

Using MAKSA technology, reproduction centers, insemination labs, breeding centers, and farmers can:

- **save around $26,000 per year** on CASA consumables (per 10,000 samples tested)
- **increase conception rates by ~30%** for their end clients
- **strengthen their reputation** and **attract more clients** through more consistent results.

Objective

To validate the smartphone-based MAKSA (Mobile-Assisted Key Semen Analysis) technology powering Avare Biotech App for F**rozen Bovine semen** evaluation against conventional expensive CASA systems

Materials and Methods

- Frozen semen samples from two bovine breeds were used in this study. The samples included Gyr semen and HF breed. Commercially prepared frozen semen straws of both species were obtained and processed for analysis.

- The semen straws were thawed according to standard laboratory procedures, after which the samples were diluted at a 1:1 ratio using Triladyl semen extender to ensure optimal sperm dispersion and motility assessment.

- Following dilution, video recordings of each semen sample were captured using a smartphone Iphone 13, following standardized recording conditions to minimize variability. At the same time, aliquots of the same samples were transferred to a separate laboratory room for evaluation using Hamilton Thorn's CASA system, ensuring parallel assessment of identical samples.

- The entire process of thawing, dilution, video recording, and CASA evaluation was completed within a time frame of approximately two hours to minimize the effects of prolonged sample handling on semen quality parameters. After completion of the analysis, the generated data from both the smartphone-based assessments and the CASA system were collected, compiled, and stored.

Table 1MAKSA results comparison

Table 1. MAKSA results comparison

Tests done	Key Semen Parameters	MAKSA	CASA	Difference %
Bull 1 (Eagle)	Concentration (M/ml)	244.4	26.95	—
	Total Motility (%)	63	61.66	2.17
	Progressive Motility (%)	55	57.51	4.36
Bull 2 (Fourtuner)	Concentration (M/ml)	367.9	27.55	—
	Total Motility (%)	69	64.8	6.48
	Progressive Motility (%)	62	63.27	2.01
Bull 3 (Thunder)	Concentration (M/ml)	289.3	25.11	--
	Total Motility (%)	67	64.73	3.51

	Progressive Motility (%)	58	60.58	4.26

Discussion / Expert Insights

- **Total and progressive motility correlates well with CASA readings**, indicating the amazing potential of the MAKSA system. Progressive motion sensitivity by the AI based algorithm works robustly which needs further calibrations.
- Concentration values vary and are dependent on the factors like dilution of the sample, relative concentration differences among the different fields of one sample.
- The system sometimes showed incompatibility with **phase contrast** microscopic recordings that is likely due to contrast differentiation between sperm and the background.
- Overall, MAKSA proved stable, reliable, and user-friendly under real-world field conditions

Recommendations

- Use bright-field microscopy only for CASA video recording; contrast/phase-contrast microscopy introduces halo artifacts and poor contrast, leading to inaccurate sperm detection and motility analysis.

- Standardize dilution, slide preparation, chamber depth, and temperature to minimize concentration variables and background fluid movement.

- To improve users' journey it's recommended to make videos inside the app under the supervision of the app and provide guided error prompts for focus, dilution, lighting, and recording mode.

Appendix

• Screenshot of the semen videos made and CASA results
• MAKSA PDF report samples

Bull 1: CASA readings

minitube

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QC Sample Report

Donor name	Eagle		
Donor ID	GYR 2231 13/03/2026	Species	Bovine
Life number		Breed	Gyr

Motility & Concentration of processed ejaculate

Analysis conditions

Sample analysis

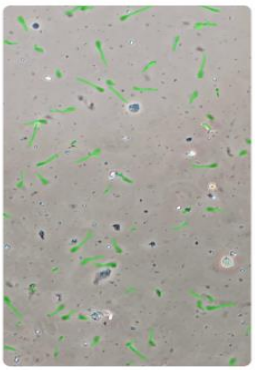
Date of analysis		13/03/2026
Date of production		13/03/2026
Total no. of sperm analysed		193.00
Number of fields		4.00
Total motility	[%]	61.66
Progressive motility	[%]	57.51
Progressive fast motility	[%]	36.79
Progressive slow motility	[%]	19.17
Progressive circular motility	[%]	1.55
Local circular motility	[%]	--
Local motility	[%]	4.15
Immotile	[%]	38.34

Notes: Concentration/Dose=26.95million
Not Valid for Court of Law

Bull 1: MAKSA values

Analysis 13.03.2026

Animal: — **Name:** Eagle **Age:** 4 **Breed:** GYR **Status:** Ready **Created:** 2026-03-13 11:35



Total objects:	81
Concentration (millions/ml^3):	244.40
Dilution rate:	[1:10]
Volume of field of view (mm^3):	
Progressive Motility:	55%
Non-Progressive Motility:	8%
Immotility:	37%

BULL 2: CASA readings

minitube

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QC Sample Report

Donor name	Fourtuner		
Donor ID	GYR 1120	Species	Bovine
Life number		Breed	HF

Motility & Concentration of processed ejaculate

Analysis conditions

Sample analysis

Date of analysis		13/03/2026
Date of production		13/03/2026
Total no. of sperm analysed		196.00
Number of fields		4.00
Total motility	[%]	64.80
Progressive motility	[%]	63.27
Progressive fast motility	[%]	42.35
Progressive slow motility	[%]	19.39
Progressive circular motility	[%]	1.53
Local circular motility	[%]	--
Local motility	[%]	1.53
Immotile	[%]	35.20

Notes: Concentration/Dose=27.55million
Not Valid for Court of Law

BULL 2: MAKSA values

Analysis 13.03.2026

Animal: — **Name:** Fortuner **Age:** 3 **Breed:** HF **Status:** Ready **Created:** 2026-03-13 11:30

Total objects:	59
Concentration (millions/ml^3):	367.90
Dilution rate:	[1:10]
Volume of field of view (mm^3):	
Progressive Motility:	62%
Non-Progressive Motility:	7%
Immotility:	31%

Agglutination

Found:	No

BULL 3: CASA readings

minitube

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QC Sample Report

Donor name	Thunder		
Donor ID	3398 20-11-2025	Species	Bovine
Life number		Breed	HF

Motility & Concentration of processed ejaculate

Analysis conditions

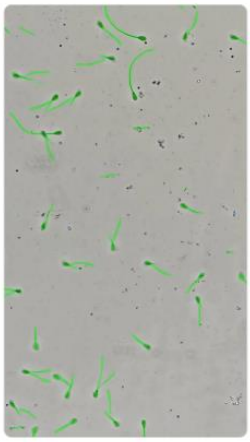
Sample analysis

Date of analysis		13/03/2026
Date of production		13/03/2026
Total no. of sperm analysed		241.00
Number of fields		4.00
Concentration	[10^6/ml]	25.11
Total motility	[%]	64.73
Progressive motility	[%]	60.58
Progressive fast motility	[%]	50.21
Progressive slow motility	[%]	9.96
Progressive circular motility	[%]	0.41
Local circular motility	[%]	--
Local motility	[%]	4.15
Immotile	[%]	35.27

BULL 3: MAKSA values

Analysis 13.03.2026

Animal: — **Name:** Thunder **Age:** 5 **Breed:** HF **Status:** Ready **Created:** 2026-03-13 11:26



Total objects:	57
Concentration (millions/ml^3):	289.30
Dilution rate:	[1:10]
Volume of field of view (mm^3):	
Progressive Motility:	58%
Non-Progressive Motility:	9%
Immotility:	33%

Agglutination

Found:	No